

Advanced Product Customization & Dynamic Pricing Rule Dashboard

I want to build an **advanced product customization and dynamic pricing rule** for a Shopify-based print/custom-product platform.

The main goal is to allow the store admin to create **flexible pricing rules from the dashboard** without changing code. Pricing rules should be configurable individually for products or collectively for collections/categories.

The system must support dimension-based, area-based, quantity-based, option-based, and combination-based pricing.

1. Product Categories

The system should support the following major product categories:

1. Apparel
2. Business Cards
3. Flyers
4. Brochures
5. Posters
6. Banners
7. Signs
8. Stickers
9. Labels
10. Mugs & Drinkware
11. Canvas Prints
12. Packaging
13. Wall Graphics
14. Window Graphics
15. Vehicle Graphics
16. Flags
17. Trade Show Displays
18. Invitations
19. Promotional Products
20. Custom Products

The system should be designed so that additional categories can easily be added later.

2. Product-Level Pricing Rules

The admin should be able to create pricing rules for an **individual product**.

Example:

Product: Vinyl Banner

The admin can configure:

- Base Price
- Minimum Price
- Width
- Height
- Thickness
- Quantity
- Area
- Material
- Finishing
- Printing Type
- Color
- Sides
- Additional Options
- Extra Charges
- Discounts

Each product can have its own completely different pricing structure.

3. Collection-Level Pricing Rules

The admin should also be able to assign a pricing rule to an entire **collection/category**.

Example:

Collection: Banners

Rule:

- Price = \$2.50 per square foot

- Minimum order = \$25
- Quantity discount:
 - 1–5 = regular price
 - 6–10 = 5% discount
 - 11–25 = 10% discount
 - 26+ = 15% discount

All products assigned to the collection should automatically use this pricing rule unless a product-specific rule overrides it.

4. Pricing Rule Priority

The system must support pricing rule priority.

Recommended priority:

1. Product-specific rule
2. Collection-specific rule
3. Category-level rule
4. Global/default rule

If a product has its own pricing rule, that rule should override the collection rule.

The admin should be able to change the priority if necessary.

5. Supported Measurement Units

The pricing engine must support multiple measurement units.

Linear Units

- Inch
- Foot
- Meter
- Centimeter
- Millimeter

Area Units

- Square Inch
- Square Foot
- Square Meter
- Square Centimeter

The admin should be able to select which unit is used for each pricing rule.

Example:

Price per square foot

Price per square meter

Price per linear foot

Price per inch

6. Dimension-Based Pricing

The system should support pricing based on:

- Width
- Height
- Length
- Thickness
- Diameter
- Area

Example:

Width = 4 ft

Height = 6 ft

Area:

$$4 \times 6 = 24 \text{ sq ft}$$

Price:

$$24 \times \$3 = \$72$$

The system should automatically calculate the final price.

7. Area-Based Pricing

Admin should be able to select:

Pricing Method → Area Based

Then configure:

- Price per square inch
- Price per square foot
- Price per square meter
- Minimum area
- Maximum area
- Minimum price

Example:

Price per square foot = \$2.50

Customer selects:

Width = 5 ft

Height = 8 ft

Area = 40 sq ft

Product price:

$40 \times \$2.50 = \100

8. Linear Pricing

The system should support linear pricing.

Example:

Banner pricing:

\$4 per linear foot

Customer selects:

Length = 10 ft

Price:

$$10 \times \$4 = \$40$$

This should also support:

- Linear inch
 - Linear foot
 - Linear meter
-

9. Base Price + Additional Price

Admin should be able to configure:

Base Price + Calculated Price + Additional Charges

Example:

Base Price = \$15

Area Price = \$2.50/sq ft

Area = 20 sq ft

Calculation:

$$\$15 + (20 \times \$2.50)$$

Final Price = \$65

10. Quantity-Based Pricing

Admin should be able to configure quantity-based pricing.

Example:

1–10 units → \$10 each

11–25 units → \$9 each

26–50 units → \$8 each
51–100 units → \$7 each
100+ units → \$6 each

The system should automatically select the correct pricing tier based on quantity.

11. Quantity Break / Volume Pricing

Support multiple quantity pricing models:

- Fixed unit price
- Tiered pricing
- Percentage discount
- Fixed discount
- Price per quantity range

Example:

Quantity	Unit Price
1–10	\$10
11–25	\$9
26–50	\$8
51–100	\$7
101+	\$6

12. Product Options & Add-On Pricing

Admin should be able to add additional pricing for product options.

Example:

Material:

- Standard Vinyl → +\$0
- Premium Vinyl → +\$10

- Reflective Vinyl → +\$25

Finishing:

- No Finishing → +\$0
- Grommets → +\$5
- Hemming → +\$8
- Pole Pocket → +\$12

Printing:

- Single-Sided → +\$0
- Double-Sided → +\$15

The system should automatically add these charges to the base calculation.

13. Conditional Pricing Rules

The pricing engine should support conditional logic.

Example:

IF Product Type = Banner
AND Material = Premium Vinyl
AND Area > 50 sq ft

THEN:

Additional Charge = \$20

Another example:

IF Quantity >= 50
THEN:

Apply 10% discount.

The admin should be able to create these conditions from the dashboard without coding.

14. Minimum & Maximum Values

For every dimension-based field, admin should be able to configure:

- Minimum value
- Maximum value
- Step/increment value
- Default value

Example:

Width:

Minimum = 1 ft
Maximum = 20 ft
Step = 0.5 ft
Default = 5 ft

Height:

Minimum = 1 ft
Maximum = 10 ft
Step = 0.5 ft
Default = 3 ft

15. Rounding Rules

The admin should be able to configure price rounding.

Options:

- No rounding
- Round up
- Round down
- Round to nearest
- Round to nearest \$0.05
- Round to nearest \$0.10
- Round to nearest \$1

Example:

Calculated Price = \$48.73

Round Up → \$49.00

16. Minimum Order Price

Admin should be able to configure a minimum product price.

Example:

Calculated price = \$12

Minimum price = \$25

Final price = \$25

If calculated price is higher than the minimum price, use the calculated price.

17. Maximum Order Limit

Admin should optionally be able to configure:

- Maximum width
- Maximum height
- Maximum area
- Maximum quantity
- Maximum order value

If the customer exceeds the allowed limit, display an appropriate validation message.

18. Custom Formula Builder

The system should include a visual formula builder.

Admin should be able to create formulas using:

- Width
- Height
- Length

- Thickness
- Area
- Quantity
- Base Price
- Unit Price
- Option Price
- Material Price
- Finishing Price
- Discount
- Markup

Supported operators:

-
-
- ×
- ÷
- %
- <
- =
- <=
- =
- AND
- OR

Example formula:

$(\text{Width} \times \text{Height} \times \text{Price Per Sq Ft}) + \text{Base Price} + \text{Finishing Charge}$

The admin should not need to write code.

19. Pricing Rule Builder UI

Create an easy-to-use dashboard interface.

Example:

Create Pricing Rule

Rule Name:

Banner Standard Pricing

Apply To:

- Product
- Collection
- Category
- All Products

Pricing Method:

Dropdown:

- Fixed Price
- Per Inch
- Per Foot
- Per Meter
- Per Sq Inch
- Per Sq Ft
- Per Sq Meter
- Quantity Based
- Dimension Based
- Formula Based

Conditions

Add Condition

[Width] [Greater Than] [10] [Feet]

AND

[Quantity] [Greater Than] [25]

Pricing

Base Price:

[\$10]

Price Per Sq Ft:

[\$2.50]

Additional Charge:

[\$5]

Discount:
[10%]

20. Live Price Preview

The admin dashboard should include a **Price Preview/Test Calculator**.

Admin can enter:

Width = 5 ft
Height = 10 ft
Quantity = 20

The system should immediately show:

Area = 50 sq ft

Base Price = \$10

Area Price = \$125

Quantity Adjustment = \$0

Additional Options = \$15

Discount = -\$10

Final Price = \$140

This allows the admin to verify pricing rules before publishing them.

21. Customer-Facing Dynamic Pricing

On the Shopify product page, the customer should see the customization fields configured by the admin.

Example:

Select Size

Width:

[5] [Feet]

Height:

[10] [Feet]

Quantity:

[20]

Material:

- ☐ Standard
- ☐ Premium

Finishing:

- ☐ None
- ☐ Grommets
- ☐ Hemming

Then dynamically display:

Estimated Price: \$140.00

The price should update instantly whenever the customer changes:

- Width
- Height
- Quantity
- Material
- Finishing
- Printing
- Any other configurable option

22. Pricing Rule Assignment

Admin should be able to assign pricing rules using:

Product

Select individual products.

Collection

Select one or multiple collections.

Category

Select product categories.

Tags

Optionally assign rules based on product tags.

Example:

Tag = **banner**

Apply Banner Pricing Rule.

23. Rule Management Dashboard

Create a dedicated:

Pricing Rules

dashboard page.

It should show:

Rule Name	Applied To	Pricing Type	Status	Priority	Actions
Banner Pricing	Banners	Per Sq Ft	Active	1	Edit
Business Card	Business Cards	Quantity	Active	2	Edit
Sticker Pricing	Stickers	Area	Draft	3	Edit

Actions:

- Create
- Edit

- Duplicate
 - Enable
 - Disable
 - Delete
 - Preview
 - Test
 - Assign
-

24. Rule Templates

Provide predefined pricing templates for common products.

Examples:

Banner Template

Width × Height × Price per Sq Ft

Sticker Template

Width × Height × Price per Sq Ft × Quantity

Business Card Template

Quantity-based pricing

Poster Template

Size + Quantity pricing

Apparel Template

Base Price + Size + Print Area + Quantity

Sign Template

Width × Height + Material + Finishing

Admin should be able to duplicate and customize these templates.

25. Multiple Pricing Rules Per Product

A product may require multiple rules.

Example:

Banner:

Rule 1 → Dimension pricing

Rule 2 → Material pricing

Rule 3 → Finishing pricing

Rule 4 → Quantity discount

Rule 5 → Rush order charge

The system should combine all applicable rules and calculate the final price correctly.

26. Extra / Rush Pricing

Admin should optionally configure additional charges such as:

- Rush Order Fee
- Design Fee
- Setup Fee
- Artwork Fee
- Customization Fee
- Shipping-related surcharge
- Minimum Order Fee

Example:

Rush Order = +20%

27. Pricing Breakdown

The customer should optionally be able to see a pricing breakdown.

Example:

Base Price: \$15.00
Size Price: \$80.00
Material: \$10.00
Finishing: \$5.00
Quantity Discount: -\$10.00

Total: \$100.00

Admin should have the option to show or hide this breakdown.

28. Admin Permissions

The system should support permissions for pricing management.

Possible roles:

- Store Owner
- Admin
- Pricing Manager
- Product Manager

Only authorized users should be able to create or modify pricing rules.

29. Important Technical Requirements

The pricing engine should be:

- Modular
- Scalable
- Fast
- Secure
- Easy to maintain
- Compatible with Shopify
- Compatible with Shopify product variants
- Compatible with Shopify cart and checkout
- Compatible with metafields where appropriate
- API-ready
- Designed for future expansion

The pricing calculation should be centralized so the same pricing logic is used consistently across:

- Product page
- Cart
- Checkout
- Admin preview
- Order data

The system must prevent customers from manipulating frontend JavaScript to bypass pricing rules.

Pricing should be validated server-side before the final cart/order price is accepted.

30. Example Use Case

Product:

Custom Vinyl Banner

Admin configures:

Pricing Method:

Per Square Foot

Base Price:

\$10

Price Per Sq Ft:

\$2.50

Minimum Price:

\$25

Maximum Width:

20 ft

Maximum Height:

10 ft

Quantity Pricing:

1–5 → Regular price
6–10 → 5% discount
11–25 → 10% discount
26+ → 15% discount

Material:

Standard → \$0
Premium → +\$10

Finishing:

None → \$0
Grommets → +\$5
Hemming → +\$8

Customer selects:

Width = 6 ft
Height = 4 ft
Quantity = 10
Material = Premium
Finishing = Grommets

Calculation:

Area = $6 \times 4 = 24$ sq ft

Area Price = $24 \times \$2.50 = \60

Base Price = \$10

Premium Material = \$10

Grommets = \$5

Subtotal = \$85

Quantity Discount = 5%

Discount = \$4.25

Final Price = \$80.75

The customer should see the updated price immediately.

31. Future-Proof Architecture

The system should be designed so that new pricing variables can be added later without rebuilding the entire pricing engine.

Future variables may include:

- Print sides
- Color count
- Ink type
- Material thickness
- Paper weight
- Finishing type
- Shape
- Cut type
- Print quality
- Production time
- Shipping zone
- Customer type
- Wholesale pricing
- Membership pricing
- Promotional pricing

The architecture should allow these variables to be added as configurable fields.

Final Goal

Build a **powerful no-code/low-code pricing rule engine** where the store administrator can manage complex pricing for all print, apparel, signage, promotional, and custom products from one dashboard.

The admin should be able to:

Create Rule → Select Product/Collection → Select Variables → Define Conditions → Create Formula → Set Pricing → Add Options → Set Quantity Rules → Set Limits → Preview Calculation → Publish

The system should automatically apply the correct pricing logic to the Shopify storefront and dynamically calculate the customer's final price based on their selected dimensions, quantity, materials, options, and customization requirements.

The overall UI should be modern, clean, intuitive, and suitable for a professional Shopify print/custom-product application.